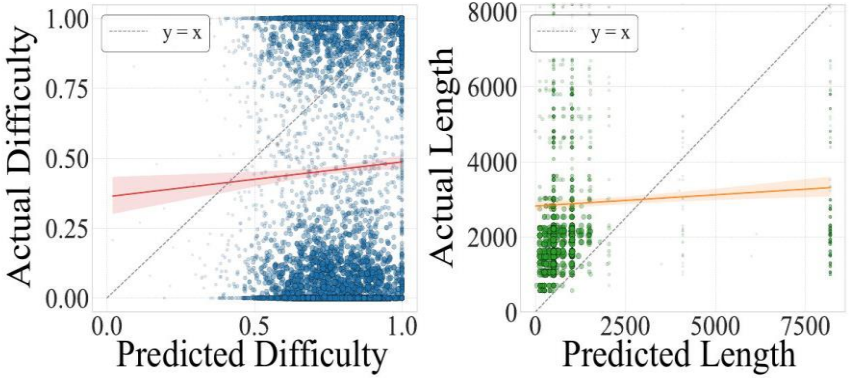
MAPR: Meta-Aware Predictive Reasoning

Hypothesis

Fostering meta-awareness of how to think directly translates to advanced reasoning performance and generalization ability.

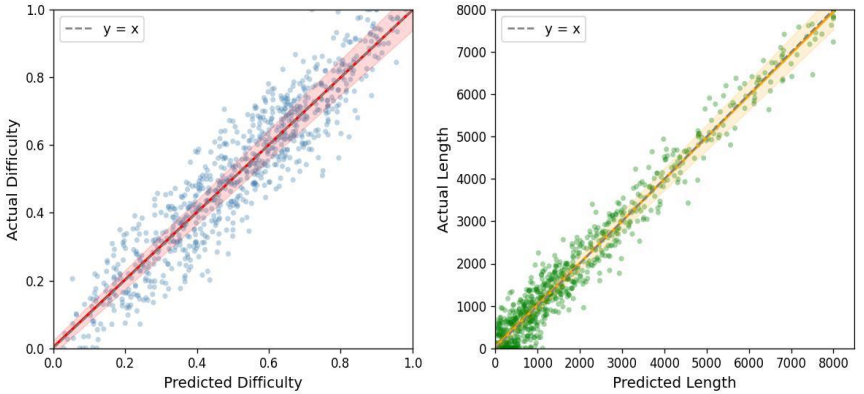
(a) GRPO — Poor Meta-Awareness



❌ **Trend lines diverge from y=x**

Predicted difficulty & length are misaligned with actual roll-out statistics — the model has no reliable self-model.

(b) MAPR — Enhanced Meta-Awareness



✅ **Trend lines tightly follow y=x**

MAPR's predictions closely track actual behavior — enabling accurate self-monitoring of difficulty and reasoning length.